Minimum Wages and the Intensive Margin: Evidence from a Stacked Event-Study of Hours Worked

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Abstract

This paper examines whether minimum wage increases generate adjustments along the intensive margin of labor input. Using IPUMS CPS data and a stacked event-study design aligned around state-level policy changes, the analysis compares workers whose pre-policy wages fall within the binding range of a minimum wage increase to higher-paid comparison workers within the same event window. The results indicate little evidence of immediate changes in weekly hours following minimum wage increases. In the medium run, estimates suggest modest reductions in hours among workers directly exposed to the policy, although statistical significance varies across exposure definitions. Overall, the findings point to limited intensive-margin adjustment in staggered minimum wage policy environments, with hours responses that are small in magnitude and concentrated among workers closest to the policy threshold.

1 Introduction

Minimum wage policy remains one of the most debated labor market interventions in the United States. While a large literature has examined employment effects, less attention has been devoted to adjustments along the intensive margin, particularly changes in hours worked among workers directly exposed to policy increases. Understanding whether hours adjust is important for assessing the broader labor-market consequences of minimum wage increases.

A substantial body of research finds that minimum wage increases generate limited aggregate employment effects, shifting attention toward alternative margins of labor market adjustment. Using geographically localized comparisons across state borders, Dube, Lester, and Reich (2016) show that minimum wage increases have little impact on employment stocks but significantly reduce labor market flows, suggesting that adjustment may occur through reduced mobility rather than immediate job loss. Related work using stacked event-study designs documents reallocation across the wage distribution, with employment declining below the new minimum wage but increasing at or slightly above it (Cengiz et al., 2019). These findings indicate that minimum wage policies can reshape labor market outcomes without producing large aggregate employment losses.

A complementary strand of research examines whether adjustment occurs through changes in hours worked. Survey-based analyses using CPS and ACS data find limited average hours responses among broad groups of low-wage workers, although reductions may occur following larger policy increases (Clemens and Strain, 2025). Other studies employ wage-based identification strategies that compare workers near the minimum wage threshold with slightly higher-paid workers, providing evidence that hours can respond for workers directly exposed to policy changes (Zavodny, 2000; Redmond and McGuinness, 2023).

Despite these advances, relatively little evidence combines dynamic event-time identification with wage-based exposure definitions to study hours responses in environments with repeated and staggered policy changes. I contribute to this literature in two ways. First, I examine the effects of state-level minimum wage increases on usual weekly hours among workers located near the policy threshold. Second, I implement a stacked event-study design covering 156 state-level minimum wage increases between 2010 and 2019, comparing workers whose pre-policy wages fall within the binding range of each policy change to workers outside the affected wage interval within the same event window. This design allows dynamic estimation of hours responses before and after each policy change while controlling for event-level and labor-market fixed effects.

The results indicate no evidence of differential pre-treatment trends. Short-run effects on weekly hours are statistically indistinguishable from zero. In the medium-run window, estimates suggest modest reductions in hours among workers classified as bound by the policy. These effects are economically small, and their statistical significance varies across exposure definitions, suggesting limited intensive-margin adjustment following minimum wage increases.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and data. Section 3 outlines the empirical strategy. Section 4 presents the main results, followed by robustness checks in Section 5. Section 6 concludes.

2 Data

2.1 Data Sources

This study combines state-level minimum wage data with individual-level labor market outcomes from the Current Population Survey (CPS). The minimum wage policy data are drawn from Vaghul and Zipperer (2019), Historical State and Sub-state Minimum Wages, Version 1.2.0. This dataset provides monthly minimum wage levels for U.S. states and selected sub-state jurisdictions.

The analysis restricts attention to state-level minimum wages and to the period January 2010 through December 2019, aligning with the time coverage of the outcome data. For each state-month, the effective minimum wage is defined as the maximum of the federal and state statutory minimum wage in that month. Individual-level outcomes and demographic characteristics are obtained from the Current Population Survey (CPS), accessed through IPUMS-CPS.

The CPS is a nationally representative monthly household survey conducted by the U.S. Census Bureau and the Bureau of Labor Statistics. The analysis uses the CPS Outgoing Rotation Group (ORG) files from 2010–2019, which contain detailed information on hourly wages and usual weekly hours worked. The CPS data are organized at the individual-by-month level, providing information on hourly wages, usual weekly hours worked, and demographic characteristics.

The combined dataset therefore links state-by-month minimum wage levels to individual-level labor market outcomes observed at monthly frequency.

2.2 Sample Construction

2.2.1 Baseline CPS Sample

The baseline sample is constructed from the monthly CPS files over the period January 2010 through December 2019. The analysis focuses on hourly-paid workers, as minimum wage changes directly bind to hourly compensation.

Individuals are therefore restricted to those who report being paid by the hour. Observations with implausible or missing values for usual weekly hours are excluded. The outcome variable, usual weekly hours worked, is drawn directly from CPS survey responses. Hourly wage information is obtained from the CPS ORG variable HOURWAGE. Observations with missing or invalid wage values are excluded. To focus on workers potentially affected by minimum wage changes, the analysis further restricts the sample to individuals whose hourly wage lies between 75% and 150% of the baseline state minimum wage.

Standard demographic controls including age, gender, education, race, and Hispanic ethnicity, are constructed using CPS variables and harmonized across years.

All analyses apply CPS sampling weights, adjusted appropriately when observations are replicated in the stacked event-study design described below, to preserve representativeness.

2.2.2 Minimum Wage Events

Minimum wage policy changes are organized into discrete events defined at the state-by-month level. An event occurs in a state-month when the state's effective minimum wage increases relative to the immediately preceding month.

Let s index states and t index calendar months. Denote the effective minimum wage in state s and month t as MW_{st}. An event is defined when $\Delta MW_{st} = MW_{st} - MW_{s,t-1} > 0$. Each event is assigned a unique identifier, e, corresponding to the treated state and the month in which the increase takes effect. Event time is indexed in relative months around the effective date of the increase.

For each event e, the policy implementation month is normalized to $\tau = 0$, and observations are indexed by $\tau \in -12, \ldots, 12$, where $\tau < 0$ denotes months before the increase and $\tau > 0$ denotes months after the increase.

The analysis focuses on this symmetric event window to capture short- and medium-run adjustments in labor supply. Because minimum wage increases may occur multiple times within a state during the sample period, event windows are constructed to avoid overlap with adjacent increases in the same treated state.

Specifically, for each treated state, the event window is truncated so that months falling on or before the previous increase, or on or after the subsequent increase, are excluded from the window for the current event. This restriction ensures that τ within each event window maps to a period in which the treated state experiences only the focal increase.

In addition, the construction of comparison states follows a "clean controls" rule. For a given event e, a state is eligible to serve as a control if it does not experience any minimum wage increase within the event window months corresponding to $\tau \in -12, \dots, 12$.

The treated state is always retained by construction, while control states are excluded if they experience an increase anywhere within the event window. This produces a set of treated and control states that are comparable in the sense that only the treated state experiences a minimum wage increase during the event window, while all states share the same calendar months for each τ. All subsequent analyses are conducted on the stacked event-study dataset constructed from these events, described next.

2.2.3 Stacked Event Panel Construction

To accommodate the staggered timing of minimum wage increases across states, the analysis employs a stacked event-study design. Each minimum wage increase is treated as a distinct policy event with its own event window. For each event, eligible treated and control states are identified, and the data are organized into event-specific panels indexed by relative event time τ. Because calendar months may fall within multiple event windows, some observations appear more than once in the stacked dataset, each time associated with a different event identifier. Sampling weights are adjusted accordingly to account for this replication. The resulting stacked event panel forms the basis for the triple-difference event-study estimation described below.

2.2.4 Treatment Definition

The empirical design exploits variation along three dimensions: event exposure, state treatment status, and pre-policy wage position relative to the minimum wage increase. Treated States For each event e, let s denote a state and τ relative event time.

Define

$$Treated_{se} = \begin{cases} 1 & \text{if state } s \text{ is the treated state for event } e, \\ 0 & \text{otherwise.} \end{cases}$$

By construction, the treated state is the state experiencing the minimum wage increase at $\tau = 0$. All other eligible states in the stacked event window serve as controls.

Defining Wage Exposure (Bound)

The second dimension of treatment captures whether a group of workers is likely to be directly affected by the minimum wage increase. Because the CPS is a repeated cross-section rather than a longitudinal panel, the same individuals cannot be followed over time. Instead, exposure to the minimum wage is defined at the level of demographic cells within each event. For each event e, individuals are grouped into demographic cells defined by combinations of state, sex, education, race, and Hispanic ethnicity.

Let c index such cells. For each cell c in state s within event e, the average pre-policy wage is computed using only months with $\tau \leq -1$:

$$\bar{w}_{cse}^{pre} = \text{weighted mean of hourly wages in cell } c \text{ for } \tau \leq -1.$$

This pre-policy average wage is used to classify whether the cell is exposed to the minimum wage increase.

Let MW_{se}^{base} denote the effective minimum wage in the treated state in the month immediately before the increase, and MW_{se}^{new} denote the new minimum wage at $\tau = 0$. The size of the increase is therefore

$$\Delta MW_{se} = MW_{se}^{new} - MW_{se}^{base}.$$

A cell is defined as "bound" if its pre-policy average wage lies within the wage interval directly affected by the increase.

To account for measurement error and heaping in reported hourly wages, and to maintain sufficient support within each stacked event window, the main specification includes a small tolerance band ε around the affected wage interval. Very small values of ε would result in some events lacking observations in either the bound or non-bound group, forcing those events to be dropped from the sample. The baseline specification sets ε = \$0.25, which balances classification accuracy with event retention. Robustness checks using alternative tolerance band definitions are presented in Section 6.1.

Formally,

$$Bound_{cse} = \begin{cases} 1 & \text{if } \bar{w}_{cse}^{pre} \in [MW_{se}^{base} - \varepsilon,\ MW_{se}^{base} + \Delta MW_{se} + \varepsilon), \\ 0 & \text{otherwise.} \end{cases}$$

Importantly, the bound classification is determined exclusively using pre-policy wages and the event-specific minimum wage change. It does not depend on post-policy outcomes.

To illustrate the definition of wage exposure, Figure 1 plots the distribution of pre-policy wages relative to the baseline minimum wage and shows how workers are classified as bound and non-bound.

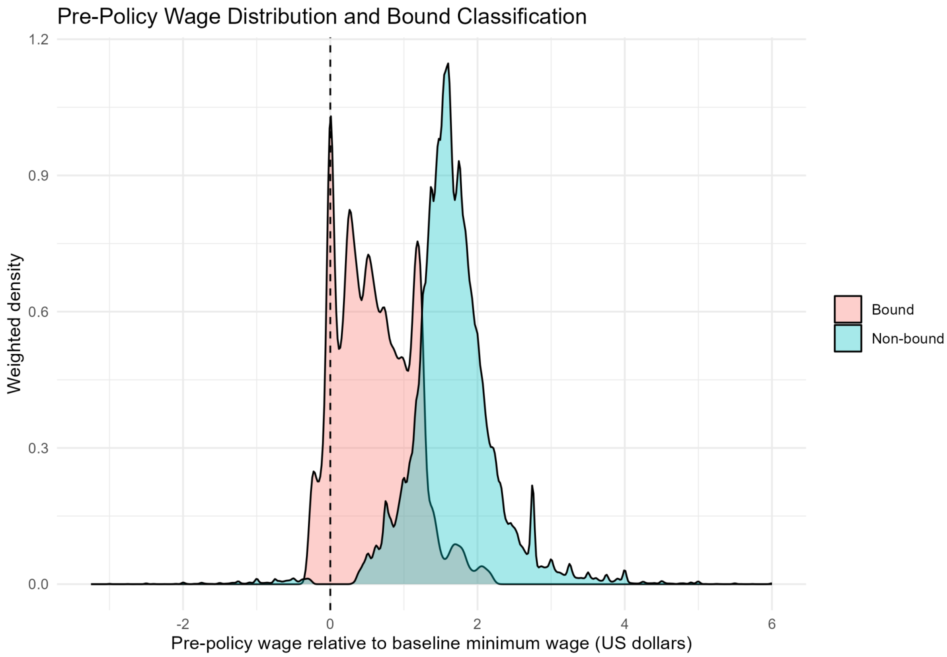

Figure 1: Pre-Policy Wage Distribution and Bound Classification

Notes: The figure plots the weighted distribution of pre-policy wages relative to the baseline minimum wage in the estimation sample. Workers classified as bound are those whose pre-policy wages fall within the event-specific wage interval used to define exposure to minimum-wage increases. Non-bound workers are those whose wages lie outside this interval. The dashed vertical line indicates the minimum wage. The figure illustrates that bound workers are concentrated near the minimum-wage threshold, while non-bound workers are located further up the wage distribution.

Bound workers are those whose pre-policy wages fall within the event-specific interval around the minimum-wage increase, while non-bound workers lie outside this range. As shown in the figure, bound workers are concentrated near the minimum-wage threshold, whereas non-bound workers are located further up the wage distribution. This pattern is consistent with the interpretation that minimum-wage increases primarily affect workers whose wages are close to the policy threshold.

Triple-Difference Structure

The estimating equation therefore compares changes in outcomes for bound versus non-bound cells in treated states relative to the same difference in control states over time. The coefficient of interest is associated with the interaction:

$$Treated_{se} \times Bound_{cse} \times \mathbf{1}\{\tau = k\}.$$

for each relative month k. Section 3.1.2 presents the full estimating equation.

This triple-difference structure isolates the differential change in labor supply for workers whose pre-policy wages place them in the affected wage interval, netting out (i) common time effects within each event window, (ii) state-level shocks, and (iii) differences between bound and non-bound cells that are constant over time.

2.3 Final Sample

The CPS ORG microdata are merged with the state-by-month minimum wage panel using state and calendar month identifiers. The merge process is explained in detail in Appendix A. The stacked event-study structure described in Section 2.2.3 is then applied, creating event-specific datasets indexed by event time relative to each policy change.

The final stacked dataset contains over three million individual-month observations covering 156 minimum wage increase events across U.S. states and the District of Columbia during 2010–2019.

Table 1 reports the final sample counts.

Table 1: Final Sample Counts

Statistic	Value
Observations (unweighted)	3,149,467
Observations (weighted)	2,018,445,876
Events	156
States	51
Treated obs (unweighted)	131,628
Control obs (unweighted)	3,017,839
Treated obs (weighted)	739,389,968
Control obs (weighted)	1,279,055,909

Notes: This table reports final sample counts for the stacked event-study sample. Because the data are stacked at the event level, a given state-month observation may appear in multiple event windows. Sampling weights are divided by the number of times each state-month appears in the stacked dataset to preserve correct aggregate representation.

2.4 Summary Statistics

Summary statistics are reported in Table 2. Bound workers work fewer hours and earn slightly lower wages than non-bound workers, consistent with their location near the minimum wage threshold.

Table 2: Summary Statistics

Group	Weekly hours	Hourly wage	Age	Share bound	Share treated	N (unweighted)	N (weighted)
Panel A: Full sample							
Full sample	30.765	9.789	34.511	0.085	0.366	3,149,467	2,018,445,876
Panel B: By bound status							
Bound	28.754	9.704	32.646	1.000	0.502	216,251	171,918,744
Non-bound	30.952	9.797	34.685	0.000	0.354	2,933,216	1,846,527,133

Notes: The table reports summary statistics for the final analysis sample. Weekly hours refers to usual weekly hours worked on the main job. Hourly wage is constructed as usual weekly earnings divided by usual weekly hours. Bound status indicates that a worker's pre-policy wage falls within the affected wage interval for a given minimum wage event. Sampling weights are applied in the weighted statistics.

3 Empirical Strategy / Methodology

3.1 Identification Framework

3.1.1 Research Design Overview

The empirical strategy is based on a stacked event-study triple-difference (DDD) design that exploits variation in the timing and magnitude of state-level minimum-wage increases between 2010 and 2019.

Each policy change defines an event window centered around the month of implementation. These event windows are stacked to construct a balanced panel of treated and comparison states within a symmetric time interval surrounding each minimum-wage increase.

The parameter of interest is the effect of minimum-wage increases on usual weekly hours worked among workers belonging to demographic cells whose pre-policy wage position places them within the interval directly affected by the policy ("bound cells"). Identification arises from three sources of variation: cross-state differences between treated and comparison states within each event window; cross-cell differences between bound and non-bound demographic cells; and within-state changes over event time relative to the month immediately preceding the policy implementation.

The identifying comparison can be summarized as follows: changes in weekly hours among bound workers in treated states after a minimum-wage increase are compared to contemporaneous changes among (i) non-bound workers within the same states and (ii) workers in comparison states that do not experience a policy change within the same event window. This triple-difference structure differences out state-level shocks common to all workers, worker-type differences that are constant over time, and aggregate time shocks within each stacked event.

3.1.2 Estimation Equation: Dynamic Event-Study Specification

The dynamic DDD specification is given by

$$
\begin{aligned}
Hours_{ise\tau} = &\sum_{k \neq -1} \beta_k \left(Bound_{ce} \times Treated_{se} \times \mathbb{1}\{\tau = k\} \right) \\
&+ \sum_{k \neq -1} \gamma_k \left(Bound_{ce} \times \mathbb{1}\{\tau = k\} \right) \\
&+ X'_{ise\tau}\delta + \alpha_{se} + \lambda_{e\tau} + \theta_{ce} + \varepsilon_{ise\tau}.
\end{aligned}
$$

where $Hours_{ise\tau}$ denotes usual weekly hours worked by individual i residing in state s during event e at event time τ, measured in months relative to the policy implementation date. The indicator $Bound_{ce}$ equals one if demographic cell c in event e is classified as being within the wage interval directly affected by the minimum-wage increase, based on its pre-policy wage position. The indicator $Treated_{se}$ equals one for states experiencing the minimum-wage increase in event e and zero for comparison states within the same stacked window. The event-time indicator $\mathbb{1}\tau = k$ captures month k relative to the policy change, with $\tau = -1$ serving as the omitted reference period.

The coefficients β_k trace the dynamic treatment effects for bound workers in treated states relative to the reference month. The sequence of β_k parameters allows the estimation of both pre-treatment dynamics and post-treatment adjustment paths. Joint Wald tests of the coefficients corresponding to pre-treatment months are used to assess the plausibility of parallel trends prior to the policy change.

The vector $X_{ise\tau}$ includes individual-level covariates such as age, age squared, union status, and indicator controls for occupation and industry. The term α_{se} denotes event-by-state fixed effects, absorbing time-invariant differences between treated and comparison states within each stacked event. The term $\lambda_{e\tau}$ represents event-by-calendar-time fixed effects, controlling for aggregate shocks common to all states within each event window. The term θ_{ce} denotes event-by-demographic-cell fixed effects, where demographic cells are defined by combinations of sex and education.

Standard errors are clustered at the state and event levels to account for serial correlation within states and dependence across observations within each stacked event window.

Identification relies on the assumption that, within each event window, trends in weekly hours for bound workers in treated states would have evolved in parallel to those of comparison groups in the absence of the minimum-wage increase. The dynamic specification provides a direct test of this assumption through the examination of pre-treatment coefficients.

3.2 Fixed Effects Structure and Controls

The empirical specification incorporates a rich set of fixed effects to isolate within-event variation and to account for potential confounding factors arising from cross-state heterogeneity, aggregate shocks, and compositional differences.

First, event-by-state fixed effects, denoted by α_{se}, absorb all time-invariant differences between treated and comparison states within each stacked event window. These fixed effects control for persistent cross-state differences in labor market conditions, institutional environments, and baseline policy regimes that may be correlated with both minimum-wage changes and labor supply outcomes.

Second, event-by-calendar-time fixed effects, denoted by $\lambda_{e\tau}$, capture shocks common to all states within a given event window at event time τ. This structure accounts for aggregate macroeconomic conditions, seasonal variation, and other time-specific factors that could affect weekly hours contemporaneously across treated and comparison states.

Third, event-by-demographic-cell fixed effects, denoted by θ_{ce}, absorb persistent differences across demographic cells within each event. Because exposure to the minimum-wage increase is defined at the cell-by-event level through $Bound_{ce}$, this set of fixed effects ensures that identification is driven by differential changes over event time rather than by baseline differences across cells.

The specification includes controls for union status, occupation, and industry as part of the control vector $X_{ise\tau}$. These controls account for persistent structural differences in typical hours worked across labor market segments. Because occupation and industry are strongly correlated with both wage levels and scheduling norms, failing to absorb these dimensions could confound estimated minimum-wage effects with compositional differences across sectors.

A potential concern is that occupation and industry may themselves adjust following minimum-wage increases. If so, conditioning on these variables could absorb part of the policy response and introduce "bad control" bias. To evaluate this possibility, the baseline specification is re-estimated without the occupation, industry, and union fixed effects in section 5.2. The point estimates remain similar in magnitude and direction across specifications, while statistical precision declines when these controls are omitted. In particular, the medium-run coefficient becomes less precisely estimated, although its sign remains unchanged.

This pattern indicates that occupation, industry, and union fixed effects primarily serve to improve precision by accounting for cross-sector heterogeneity in baseline hours, rather than mechanically absorbing the policy-induced response. The stability of point estimates across specifications suggests that the main findings are not driven by conditioning on post-treatment adjustment margins.

Taken together, the fixed-effects structure ensures that identification of the dynamic treatment coefficients β_k relies on within-event, within-state, and within-cell variation over event time, net of persistent cross-sectional heterogeneity and common shocks. Under the maintained parallel-trends assumption within each stacked window, the remaining variation isolates the differential evolution of weekly hours for bound cells in treated states relative to appropriate comparison groups.

3.3 Heterogeneity by Policy Intensity

While the baseline specification estimates the average dynamic effect of minimum-wage increases across all events, policy changes differ substantially in magnitude. To examine whether labor supply responses vary with the size of the statutory increase, three complementary approaches are implemented. These approaches distinguish between discrete group differences, linear intensity effects, and event-level heterogeneity, without imposing a priori functional-form restrictions on the relationship between policy size and labor supply adjustments.

3.3.1 Continuous Dose-Response Specification

The first approach evaluates whether the labor supply response varies smoothly with the magnitude of the policy change. Rather than partitioning events into discrete bins, the continuous value of ΔMW_e is interacted with the post-treatment DDD indicator.

Specifically, the following augmented specification is estimated:

$$\begin{aligned} Hours_{ise\tau} = {} & \beta_1 \left(Post_\tau \times Bound_{ce} \times Treated_{se}\right) \\ & + \beta_2 \left(Post_\tau \times Bound_{ce} \times Treated_{se} \times \Delta MW_e\right) \\ & + X'_{ise\tau}\delta + \alpha_{se} + \lambda_{e\tau} + \varepsilon_{ise\tau}. \end{aligned}$$

where $Post_\tau$ denotes indicators for post-policy periods. The coefficient β_2 captures a linear dose-response relationship between the size of the minimum-wage increase and the post-treatment change in weekly hours for bound cells in treated states. A statistically significant β_2 would indicate that larger policy changes are associated with proportionally larger labor supply adjustments.

3.3.2 Discrete Shock Groups

The second approach classifies events into discrete groups based on the magnitude of the event-level minimum-wage increase, denoted by ΔMW_e. Events are partitioned into terciles of the empirical distribution of ΔMW_e, generating small-, medium-, and large-increase categories.

This grouping allows the analysis to examine whether labor supply responses differ between modest minimum-wage adjustments and larger policy shocks. In the sample, small shocks correspond to very small statutory adjustments of only a few cents (around \$0.04 \$0.05), while medium shocks are typically increases of roughly \$0.35–\$0.40. Large shocks represent substantially larger step increases, often exceeding one dollar in a single adjustment, for example, the increases implemented in Arizona and Washington in 2017.

The baseline DDD specification is then estimated separately within each shock group.

Formally, for each subgroup $g \in$ small, medium, large, the following specification is estimated:

$$Hours_{ise\tau} = \sum_{k \neq -1} \beta_k^{(g)} \left(Bound_{ce} \times Treated_{se} \times \mathbb{1}\{\tau = k\} \right)$$

$$+ X'_{ise\tau}\delta + \alpha_{se} + \lambda_{e\tau} + \theta_{ce} + \varepsilon_{ise\tau}.$$

This strategy allows the dynamic response coefficients $\beta_k^{(g)}$ to vary across policy-intensity groups, thereby testing whether adjustments in weekly hours are concentrated among larger statutory increases.

3.3.3 Event-Level Heterogeneity

The third approach avoids imposing either discrete bins or linear functional forms. Instead, event-specific treatment effects are estimated by allowing the post-treatment DDD coefficient to vary across events. This is implemented by interacting the post-treatment bound-by-treated indicator with event fixed effects:

$$Hours_{ise\tau} = \sum_{e} \theta_e^{mid} \left(\mathbb{1}\{e\} \times Post_\tau^{mid} \times Bound_{ce} \times Treated_{se} \right)$$

$$+ X'_{ise\tau}\delta + \alpha_{se} + \lambda_{e\tau} + \theta_{ce} + \varepsilon_{ise\tau}.$$

The event-specific coefficients θ_e^{mid} are then examined as a function of ΔMW_e through graphical analysis. This strategy provides a non-parametric assessment of whether treatment effects are concentrated in the upper tail of the policy-size distribution and whether the relationship exhibits discontinuities or gradual patterns.

Together, these three approaches allow for a systematic evaluation of intensity heterogeneity without relying exclusively on a single functional-form assumption. The analysis therefore distinguishes between average effects, smooth dose-response relationships, and potential right-tail concentration of treatment effects.

Having outlined the identification strategy and estimation framework, the next section applies this stacked event-study design to CPS data to estimate the effects of minimum wage increases on weekly hours worked.

4 Results

The analysis is conducted on a symmetric event window centered around the policy change, with $\tau \in [-6, 12]$ used for estimation. Although the underlying stacked dataset is constructed over a broader horizon, restricting the main specification to the six months prior to treatment focuses identification on the period closest to the policy change, where event support is stable and violations of parallel trends would be most consequential. Pre-trends are therefore assessed over $\tau = -6$ to $\tau = -2$, emphasizing near-term dynamics rather than distant pre-treatment months that may be influenced by unrelated shocks or limited support.

Inspection of event-level support further reveals that the number of contributing events declines sharply at $\tau = 12$, where only a small subset of events remain in the stacked sample relative to earlier post-treatment months. Because estimates at this horizon are based on substantially reduced event support, including $\tau = 12$ in the medium-run bin risks allowing a sparsely supported month to exert disproportionate influence on joint tests. For this reason, the medium-run window is defined as $\tau = 7$ to $\tau = 11$, ensuring that bin estimates reflect periods with broad and stable event support. Appendix Table B reports the distribution of event support across event time.

4.1 Baseline Dynamic Effects

Figure 2 presents the dynamic event-study estimates of the effect of minimum-wage increases on usual weekly hours worked among bound workers. The coefficients trace the estimated path of β_k from six months prior to policy implementation through twelve months after, with $\tau = -1$ serving as the omitted reference period.

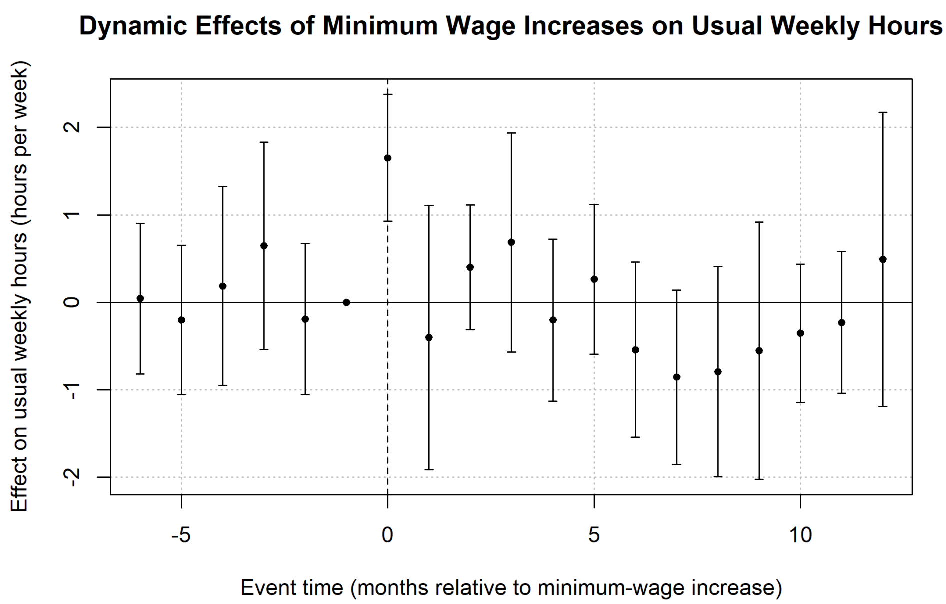

Figure 2: Baseline event-study estimates

Notes: The figure plots stacked event-study estimates of the treated × bound effect on usual weekly hours worked. Each point corresponds to the coefficient on the interaction between event time τ and the treated-bound indicator, with $\tau = -1$ omitted as the reference period. Vertical bars denote 95% confidence intervals with standard errors clustered at the state and event levels.

Across the pre-policy window, the estimated coefficients are small in magnitude and statistically indistinguishable from zero. A joint Wald test over $\tau \in [-6, -2]$ fails to reject the null hypothesis of no differential pre-trend (p = 0.81), providing support for the identifying assumption of parallel trends.

Table 4.1 Baseline Bin Estimates and Alternative Clustering Structures			
	Baseline (State & Event)	State only	Event only
Post (1–6) × Treated × Bound	0.107 (0.243)	0.107 (0.240)	0.107 (0.179)
Post (7–11) × Treated × Bound	-0.463* (0.249)	-0.463* (0.239)	-0.463** (0.205)
Pre-trend Wald p-value (τ = −6 to −2)	0.809	0.769	0.722
Observations	2,958,553	2,958,553	2,958,553
Event × State FE	Yes	Yes	Yes
Event × Month FE	Yes	Yes	Yes
Event × Cell FE	Yes	Yes	Yes
Industry / Occupation / Union FE	Yes	Yes	Yes
Clustering	Two-way: state & event	One-way: state	One-way: event

Notes: The dependent variable is usual weekly hours worked. The reported coefficients correspond to the interaction between post-treatment indicators, treated status, and bound-worker status in a stacked triple-difference specification. The post (1–6) and post (7–11) indicators correspond to short-run and medium-run post-treatment periods relative to the minimum-wage change event. All specifications include event-by-state, event-by-month, and event-by-cell fixed effects, as well as industry, occupation, and union fixed effects. Regressions are weighted using adjusted CPS earnings weights. Standard errors are clustered as indicated in the column headers. The sample consists of CPS MORG workers in the analysis sample. Bound workers are defined based on baseline wages relative to the minimum wage. Pre-trend Wald tests are reported for leads from event time −6 to −2. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

In the short-run window ($\tau = 1$ to 6), the joint post-treatment effects are statistically indistinguishable from zero (p = 0.42), indicating no immediate adjustment in usual weekly hours following minimum-wage increases.

In the medium-run window ($\tau = 7$–11), the joint test yields statistical significance at 10% level (p = 0.068). The corresponding bin estimates in Table 4.1 indicate a medium-run reduction of approximately 0.46 hours per week under the baseline two-way clustering scheme, with similar point estimates under alternative clustering choices. However, this effect is modest in magnitude and not uniformly distributed across individual event-time coefficients.

Taken together, the baseline specification indicates limited evidence of hours adjustment following minimum-wage increases. While a moderate negative effect appears in the medium-term window, the overall pattern suggests that any intensive-margin response is small and temporally localized rather than large or persistent.

4.2 Dose–Response Specification

To examine whether the effects of minimum-wage increases scale proportionally with the size of the policy change, a continuous dose–response specification is estimated. Specifically, the baseline bin indicators are augmented with interactions between the treatment term and the event-level minimum-wage increase, ΔMW_e, yielding the following additional components: $\beta_2(Post_\tau \times Bound_{ce} \times Treated_{se} \times \Delta MW_e)$

In this specification, the coefficient on the interaction with ΔMW_e captures whether larger statutory increases generate systematically larger changes in hours among bound workers.

Table 4.2 reports the results.

Table 4.2 Dose–Response Test: Interaction with Minimum Wage Increase

	Baseline	Dose–Response
Post (1–6) × Treated × Bound	0.107 (0.243)	-0.206 (0.376)
Post (7–11) × Treated × Bound	-0.463* (0.249)	-0.418 (0.566)
Post (1–6) × Treated × Bound × ΔMW	—	0.429 (0.384)
Post (7–11) × Treated × Bound × ΔMW	—	-0.039 (0.530)
Observations	2,958,553	2,958,553
R-squared	0.289	0.289
Event × State FE	Yes	Yes
Event × Month FE	Yes	Yes
Event × Cell FE	Yes	Yes
Industry / Occupation / Union FE	Yes	Yes
Clustered SE	State & Event	State & Event

Notes: The dependent variable is usual weekly hours worked. The baseline specification corresponds to the stacked triple-difference model comparing treated and control states, bound and non-bound workers, before and after minimum-wage changes. The dose–response specification interacts the post-treatment treated × bound indicator with the size of the minimum-wage increase (ΔMW) to test whether treatment effects vary with policy intensity. Post (1–6) and Post (7–11) correspond to short-run and medium-run post-treatment periods, respectively. All regressions include event-by-state, event-by-month, and event-by-cell fixed effects, as well as industry, occupation, and union fixed effects. Regressions are weighted using adjusted CPS earnings weights. Standard errors are clustered at the state and event level. The sample consists of CPS ORG workers in the analysis sample. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

The baseline post-treatment coefficients remain statistically indistinguishable from zero in both the short-run (τ = 1–6) and medium-run (τ = 7–11) windows. More importantly, the interaction terms with ΔMW_e are also small and statistically insignificant. In the short-run window, the estimated slope with respect to policy size is 0.429 (s.e. = 0.384), while in the medium-run window the estimate is −0.039 (s.e. = 0.530). Neither coefficient is statistically different from zero at conventional levels.

These results provide no evidence of a linear dose–response relationship between the magnitude of minimum-wage increases and subsequent changes in hours among bound workers. The absence of statistically significant interaction effects suggests that the estimated impacts do not increase proportionally with the size of the statutory wage adjustment within the observed range of policy variation.

Taken together with the baseline event-study results, the continuous specification indicates that average hours responses do not exhibit a smooth monotonic relationship with policy intensity.

4.3 Heterogeneity by Policy Intensity (Terciles)

To explore whether potential non-linear responses are masked in the continuous specification, events are partitioned into terciles based on the distribution of the statutory increase ΔMW_e. The baseline bin specification is then estimated separately for the small, medium, and large increase groups.

Table 4.3 reports the results.

Table 4.3 Heterogeneity by Policy Intensity (Terciles)

	Small	Medium	Large
Post (1–6) × Treated × Bound	-0.189 (0.541)	0.216 (0.444)	0.143 (0.197)
Post (7–11) × Treated × Bound	-1.090 (1.007)	-0.236 (0.404)	-0.341* (0.201)
Pre-trend Wald p-value (τ = −6 to −2)	0.091	0.908	0.122
Pre-trend Wald p-value (τ = −4 to −2)	0.990	0.804	0.405
Events	52	56	48
Observations	1,132,188	1,053,939	772,413
R-squared	0.286	0.294	0.298
Event × State FE	Yes	Yes	Yes
Event × Month FE	Yes	Yes	Yes
Event × Cell FE	Yes	Yes	Yes
Industry / Occupation / Union FE	Yes	Yes	Yes
Clustered SE	State & Event	State & Event	State & Event

Notes: The dependent variable is usual weekly hours worked. The table reports stacked triple-difference estimates separately by terciles of event-level minimum-wage increase size (policy intensity), classified as small, medium, and large minimum-wage changes. The reported coefficients correspond to the interaction between post-treatment indicators, treated status, and bound-worker status within each policy-intensity group. Post (1–6) and Post (7–11) correspond to short-run and medium-run post-treatment periods, respectively. The row labeled "Events" reports the number of distinct minimum-wage policy events in each tercile, while "Observations" reports the number of stacked worker-event-month observations used in estimation. All regressions include event-by-state, event-by-month, and event-by-cell fixed effects, as well as industry, occupation, and union controls. Regressions are weighted using adjusted CPS earnings weights. Standard errors are clustered at the state and event level. Pre-trend Wald tests are reported for leads from event time −6 to −2 and −4 to −2. The sample consists of CPS ORG workers in the analysis sample. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

In the short-run window ($\tau = 1$–6), none of the tercile groups exhibits statistically significant effects. Estimated coefficients are small in magnitude and imprecisely estimated across all three groups.

In the medium-run window ($\tau = 7$–11), the small and medium increase groups do not display statistically significant effects. For the large increase group, the estimated coefficient is −0.341 (s.e. = 0.201), statistically significant at the 10 percent level. This estimate corresponds to a modest reduction in usual weekly hours among bound workers in treated states following relatively large statutory increases.

Several features of the tercile results qualify this pattern. Statistical significance appears only for the large-increase group and only in the medium-run window. The pre-trend Wald test over $\tau \in [-6, -2]$ yields a p-value of 0.091 for the small-increase subsample, indicating weaker support for parallel trends in that group. When the pre-period is restricted to $\tau \in [-4, -2]$, p-values increase for both the small and large subsamples, indicating that deviations are concentrated in the more distant pre-treatment months. Dividing events into terciles also reduces the number of treated observations within each subsample, resulting in larger standard errors and lower statistical precision.

Overall, the tercile specification indicates that any hours reductions are concentrated among larger minimum-wage increases and arise in the medium-run window. The absence of short-run effects and the limited precision within subsamples imply that heterogeneity by policy intensity is modest rather than systematic.

4.4 Event-Level Visualization

To further assess whether the tercile results reflect a sharp threshold effect, Figure 3 plots event-specific medium-run treatment effects against the size of the statutory increase, ΔMW_e.

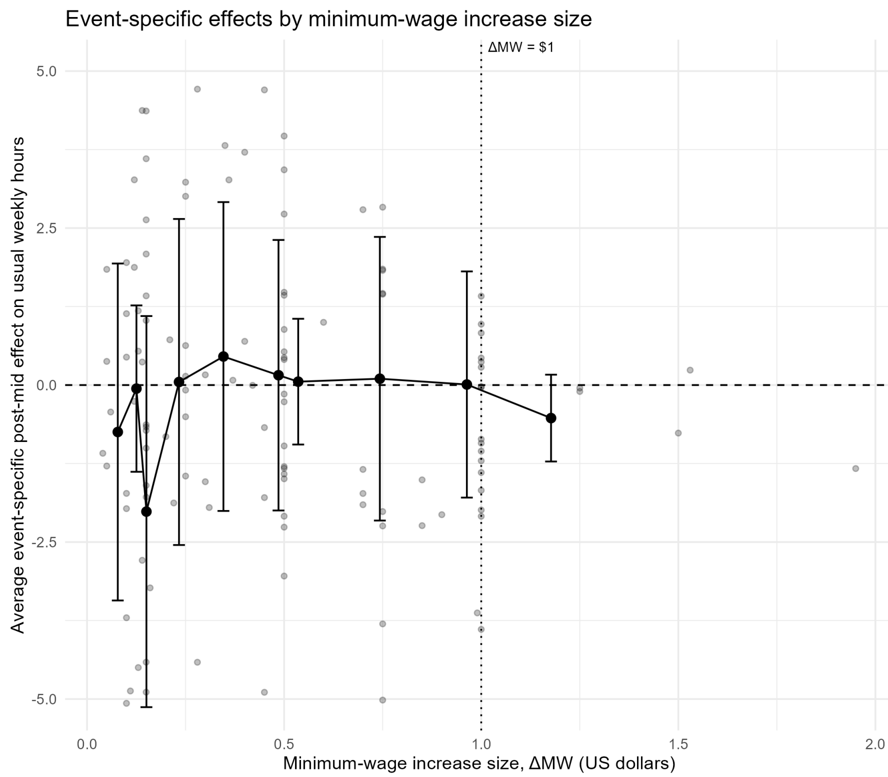

Figure 3: Event-specific post-mid effects plotted against the size of the minimum-wage increase (ΔMW)

Notes: The figure plots event-specific post-mid effects from the stacked triple-difference specification against the size of the minimum-wage increase. Each gray dot represents an event-level estimate, while black markers show binned averages with 95% confidence intervals. There are 145 events in total, and event support is reported in Appendix C. Regressions include event-by-state, event-by-time, and event-by-cell fixed effects and are weighted by CPS earnings weights. Standard errors are clustered by state and event. Bound workers are defined based on baseline wages relative to the minimum wage, and the post-mid period corresponds to the medium-run post-treatment window.

Across the distribution of policy changes, the relationship appears broadly flat, with substantial dispersion in event-level estimates. While larger policy increases are occasionally associated with more negative hours effects, the scatter does not reveal a clear monotonic relationship between policy size and the magnitude of the estimated response.

Overall, the figure provides little evidence of a sharp threshold or nonlinear discontinuity in hours responses by policy size.

4.5 Main Empirical Findings

The empirical results present a consistent pattern across specifications. The stacked event-study estimates indicate no immediate adjustment in weekly hours following minimum-wage increases. In the medium-run window, however, bound workers exhibit a modest reduction in hours relative to comparison groups, with estimated effects of roughly 0.4–0.5 hours per week.

Additional specifications examining heterogeneity by the size of the statutory increase provide limited evidence of systematic dose–response effects. Linear interactions with policy size are statistically insignificant, and partitioning events into terciles reveals significant reductions only for the largest increases.

Overall, the results suggest that minimum-wage increases during the sample period are not associated with large or persistent reductions in hours among workers near the policy threshold. Where negative effects appear, they are modest in magnitude and concentrated in the medium-run period, suggesting that hours adjustments are limited and do not scale mechanically with the size of minimum-wage increases.

5 Robustness Check

5.1 Bound Definition and Event Support

A central component of the identification strategy is the definition of exposure to the minimum-wage increase. In the baseline specification, a demographic cell c in state s and event e is classified as bound if its pre-policy average wage satisfies $\bar{w}^{pre}_{cse} \in [MW^{base}_{se} - \varepsilon,\ MW^{base}_{se} + \Delta MW_{se} + \varepsilon)$.

where $MW^{base}se$ denotes the effective minimum wage in the treated state immediately prior to the increase, $\Delta MW se$ is the size of the statutory change, and ε is a tolerance parameter introduced to account for measurement error and limited support near the policy threshold.

The choice of ε affects both the composition of the bound group and the extent of event support. When $\varepsilon = 0$, the bound interval coincides exactly with the mechanically affected wage range. In this case, some treated events contain no bound observations in the treated state within the estimation window, implying that the interaction term $Treated_{se} \times Bound_{cse}$ is identically zero for those events. Such events contribute no identifying variation for the triple-difference parameter. To ensure that identification is based on events with positive treated-side exposure, events for which the treated-state bound share equals zero are excluded in all specifications.

To assess sensitivity to the exposure definition, the model is re-estimated under alternative tolerance values $\varepsilon \in 0, ; 0.25, ; 0.50$. For each value, events with zero treated-side bound share are dropped, and the dynamic and binned triple-difference specifications are re-estimated using the same stacked window $\tau \in [-6, 12]$, excluding $\tau = 0$.

5.1 Robustness: Bound Definition and Event Dropping

	eps=0	eps=0.25	eps=0.50
Post (1–6) × Treated × Bound	0.116 (0.319)	0.108 (0.243)	0.249* (0.146)
Post (7–11) × Treated × Bound	-0.407** (0.174)	-0.463* (0.249)	-0.270 (0.220)
Pre-trend Wald p-value (τ = −6 to −2)	0.371	0.813	0.112
Treated events dropped (share_bound = 0)	18	1	0
Event × State FE	Yes	Yes	Yes
Event × Month FE	Yes	Yes	Yes
Event × Cell FE	Yes	Yes	Yes
Industry / Occupation / Union FE	Yes	Yes	Yes
Age Controls	Age, Age^2	Age, Age^2	Age, Age^2
Clustered SE	State & Event	State & Event	State & Event

Notes: The dependent variable is usual weekly hours worked. The table reports stacked triple-difference estimates using alternative definitions of bound workers based on an earnings distance threshold ε relative to the minimum wage. Workers are classified as bound if their baseline wage lies within ε dollars of the minimum wage. Columns vary ε = 0, 0.25, and 0.50. When ε = 0, treated events with no bound workers (share_bound = 0) are dropped; the number of dropped treated events is reported in the table. Post (1–6) and Post (7–11) correspond to short-run and medium-run post-treatment periods, respectively. All regressions include event-by-state, event-by-month, and event-by-cell fixed effects, as well as industry, occupation, and union fixed effects, and age and age-squared controls. Regressions are weighted using adjusted CPS earnings weights. Standard errors are clustered at the state and event level. The sample consists of CPS ORG workers in the analysis sample. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Across specifications, joint pre-treatment Wald tests over $\tau \in [-6, -2]$ fail to reject the null of no differential pre-trend. Pre-trend p-values remain above conventional significance thresholds for all values of ε, indicating that the parallel-trends diagnostics are not driven by the choice of exposure tolerance.

In the short-run window ($\tau = 1$–6), estimated coefficients are small in magnitude across specifications. Only the $\varepsilon = 0.50$ definition yields marginal significance at the 10 percent level, and the point estimates remain economically modest.

In the medium-run window ($\tau = 7$–11), the estimated coefficient is negative for all values of ε. The magnitude is largest at $\varepsilon = 0.25$, smaller at $\varepsilon = 0.50$, and remains negative when $\varepsilon = 0$. Statistical significance is strongest under the smallest tolerance and attenuates as the bound interval expands.

Expanding ε includes workers further from the minimum wage threshold, which may dilute estimated effects if adjustments are concentrated among workers very close to the policy cutoff.

Overall, the robustness exercise yields three conclusions. First, the qualitative pattern of coefficients is stable across alternative exposure definitions. Second, the absence of short-run effects does not depend on the tolerance choice. Third, medium-run reductions in hours are present across specifications but are moderate in magnitude and attenuate as the exposure definition broadens, indicating that any adjustment is concentrated among workers closest to the statutory threshold rather than diffuse across the lower-wage distribution.

5.2 Inclusion of Industry, Occupation, and Union Fixed Effects

An additional robustness check re-estimates the baseline specification without the industry, occupation, and union fixed effects discussed in Section 3.3.

5.2 Robustness: Dropping Industry/Occupation/Union Controls		
	Full controls	Drop OCC/IND/UNION
Post (1–6) × Treated × Bound	0.107 (0.243)	0.074 (0.247)
Post (7–11) × Treated × Bound	-0.463* (0.249)	-0.355 (0.233)
Pre-trend Wald p-value (τ = −6 to −2)	0.809	0.744
Observations	2,958,553	2,958,554
Event × State FE	Yes	Yes
Event × Month FE	Yes	Yes
Event × Cell FE	Yes	Yes
Industry / Occupation / Union FE	Yes	No
Clustered SE	State & Event	State & Event

Notes: The dependent variable is usual weekly hours worked. The table reports stacked triple-difference estimates comparing treated and control states, bound and non-bound workers, before and after minimum-wage changes. The first column includes the full set of industry, occupation, and union fixed effects, while the second column drops these controls to assess robustness to their inclusion. Post (1–6) and Post (7–11) correspond to short-run and medium-run post-treatment periods, respectively. All regressions include event-by-state, event-by-month, and event-by-cell fixed effects. Regressions are weighted using CPS earnings weights. Standard errors are clustered at the state and event level. The sample consists of CPS MORG workers in the analysis sample. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

When these controls are included, the medium-run bin coefficient for $\tau = 7$–11 is negative and marginally statistically significant. Dropping the controls reduces the magnitude of the estimate and eliminates statistical significance, although the coefficient remains negative. The short-run bin ($\tau =$ 1–6) remains small and statistically indistinguishable from zero under both specifications.

Pre-trend diagnostics are stable across specifications. The Wald test for joint significance of the pre-period coefficients over $\tau = -6$ to -2 yields large p-values in both models, indicating no evidence of differential pre-treatment trends.

These results suggest that the estimated medium-run reduction in hours is not driven by conditioning on industry, occupation, or union controls, although these variables appear to improve statistical precision.

5.3 Summary of Robustness Checks

The robustness exercises indicate that the main findings are qualitatively stable across alternative specifications. Varying the tolerance parameter and event-dropping rules affects statistical significance but does not alter the direction of the estimated effects. Similarly, removing industry, occupation, and union fixed effects reduces precision but leaves point estimates largely unchanged.

Overall, the robustness checks suggest that the estimated reductions in hours are modest, and their statistical significance varies across exposure definitions, but their directional stability across models is consistent with a small adjustment among workers near the minimum-wage threshold.

6 Conclusion

This paper studies whether minimum wage increases generate adjustments along the intensive margin of labor input. Using CPS ORG microdata and a stacked event-study design aligned around state-level policy changes, the analysis compares workers whose pre-policy wages fall within the binding range of each increase to workers outside the affected wage interval.

Across specifications, the results provide little evidence of large reductions in weekly hours following minimum wage increases. Dynamic estimates show no immediate adjustment in the short run, while the medium-run window exhibits modest reductions in hours among workers classified as bound by the policy. These effects are economically small and remain directionally stable across alternative exposure definitions and specification choices.

These findings complement existing evidence suggesting that minimum wage increases often reshape the wage distribution without generating substantial employment losses. In particular, the results are consistent with studies documenting limited employment effects and modest hours adjustments among low-wage workers (Clemens & Strain, 2025), as well as with evidence that minimum wage increases primarily reallocate employment across wage bins rather than inducing large aggregate contractions

(Cengiz et al., 2019).

One possible interpretation is that firms adjust gradually along scheduling margins rather than through large immediate reductions in labor demand. Employers may respond to higher wage floors by modestly reducing hours for workers closest to the statutory threshold or by adjusting scheduling practices over time. At the same time, workers may adjust labor supply in response to higher wages or changing job opportunities. Because the present analysis cannot separately identify employer-driven scheduling responses from worker-side labor supply adjustments, these mechanisms remain an important topic for future research.

Overall, the evidence suggests that minimum wage increases during the sample period were associated with modest and gradual adjustments in hours among workers most directly exposed to the policy threshold, rather than large or pervasive contractions in labor input.

References

Cengiz, D., Dube, A., Lindner, A., & Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. *The Quarterly Journal of Economics, 134*(3), 1405–1454. https://doi.org/10.1093/qje/qjz014

Clemens, J., & Strain, M. R. (2025). *The heterogeneous effects of large and small minimum wage changes on hours worked: Evidence using a partially pre-committed analysis plan* (IZA Discussion Paper No. 17913). Institute of Labor Economics (IZA). https://www.aei.org/research-products/working-paper/the-heterogeneous-effects-of-large-and-small-minimum-wage-changes-on-hours-worked-evidence-using-a-partially-pre-committed-analysis-plan/

Dube, A., Lester, T. W., & Reich, M. (2016). Minimum wage shocks, employment flows, and labor market frictions. *Journal of Labor Economics, 34*(3), 663–704. https://doi.org/10.1086/685449

Flood, S., King, M., Rodgers, R., Ruggles, S., Warren, J. R., et al. (2025). *IPUMS CPS (Version 13.0)* [Data set]. IPUMS. https://doi.org/10.18128/D030.V13.0

Redmond, P., & McGuinness, S. (2025). The impact of a minimum wage increase on hours worked: Heterogeneous effects by gender and sector. *Economica, 92*(365), 84–106. https://doi.org/10.1111/ecca.12555

Vaghul, K., & Zipperer, B. (2019). *Historical state and sub-state minimum wages (Version 1.2.0)* [Data set]. https://github.com/benzipperer/historicalminwage/releases/tag/v1.2.0

Zavodny, M. (2000). The effect of the minimum wage on employment and hours. *Labour Economics, 7*(6), 729–750. https://doi.org/10.1016/S0927-5371(00)00021-X

Appendix

Appendix A: Data Merging and Event Construction

The construction of the final analytical dataset proceeds in three stages.

First, the individual-level CPS microdata are merged with the state-by-month minimum wage panel. Each CPS observation is matched to the effective minimum wage in the corresponding state and calendar month using state and time identifiers. This merge assigns to each individual the contemporaneous statutory minimum wage applicable in their state of residence.

Second, the stacked event structure described in Section 2.3 is applied. For each minimum wage increase event, eligible treated and control states are identified within the event window. The CPS microdata are then merged with the event-specific state-by-month panel, expanding the dataset so that each observation is indexed by $(event_id, s, \tau)$. Because calendar months may belong to multiple event windows, some state-month observations appear multiple times in the stacked dataset, each time associated with a different event identifier.

Third, event-level minimum wage information is merged into the stacked dataset. For each event e, the baseline minimum wage MW_{se}^{base} and the new minimum wage MW_{se}^{new} at $\tau = 0$ are attached to all observations belonging to that event. These event-specific values are used to compute the size of the increase, ΔMW_{se}, and to define the bound classification described in Section 2.4. Throughout this process, CPS sampling weights are adjusted to account for replication induced by stacking.

Specifically, the original sampling weight for each state-month observation is divided by the number of times that state-month appears across events, ensuring that states with more frequent policy changes do not receive disproportionate influence in the estimation.

The resulting dataset is an individual-level stacked event panel in which each observation is uniquely indexed by $(event_id, s, \tau, c)$, where c denotes the demographic cell used to define exposure.

Appendix B: the number of distinct policy events contributing observations at each event time

Appendix Table B reports the number of distinct policy events contributing observations at each event time τ within the main estimation window. The table presents three measures: (i) the number of distinct events with any observations at a given τ, (ii) the number of distinct events contributing treated observations, and (iii) the total number of treated observations at that horizon.

Appendix B: Event Support by Event Time (τ)

τ	Events (any obs)	Events (treated obs)	Treated obs
-6	154	154	8,265
-5	156	156	8,309
-4	156	156	8,400
-3	156	156	8,179
-2	156	156	7,870
-1	156	156	8,081
0	156	156	7,189
1	156	156	6,783
2	156	156	6,692
3	155	155	6,979
4	155	155	6,850
5	155	155	6,837
6	150	150	6,823
7	149	149	6,798
8	149	149	6,854
9	147	147	6,657
10	147	147	6,412
11	147	147	6,569
12	23	23	1,081

Notes: This table reports the number of policy events contributing observations at each event time in the stacked event-study sample. "Events (any obs)" counts the number of events with at least one observation in the stacked panel at event time τ, while "Events (treated obs)" counts the number of events with at least one treated observation at that event time. Event support declines at longer horizons because not all policy events have complete event windows within the sample period, particularly for late policy changes near the end of the sample. Therefore, long-run estimates are based on a smaller subset of events.

Event support is broadly stable from $\tau = -6$ through $\tau = 11$, with approximately 147–156 distinct events contributing treated observations at each month. In contrast, support declines sharply at $\tau = 12$, where only 23 events remain in the stacked sample. This discontinuous reduction reflects the mechanical structure of staggered adoption combined with the finite panel horizon.

Because estimates at $\tau = 12$ are based on a substantially narrower subset of events, including this month in post-treatment bins would allow a sparsely supported horizon to exert disproportionate influence on joint tests. For this reason, the medium-run bin in the main text is defined over $\tau = 7$ to $\tau = 11$, ensuring that bin estimates reflect periods with broad and stable event support.

Appendix C: Event support for event-specific post-mid estimates

Table 3: Appendix Table C. Event support for event-specific post-mid estimates

Policy intensity group	Total events	Events with estimated post-mid effect	Dropped events	Dropped share
Small	52	47	5	9.6%
Medium	56	53	3	5.4%
Large	48	45	3	6.2%
Total	156	145	11	7.1%

Notes: The table reports event support for the event-specific medium-run estimates used in Figure 3. Events are classified into terciles based on the event-level minimum-wage increase. An event is counted as estimated if the event-specific post-mid coefficient is identified and reported in the event-level regression. Dropped events are events for which the post-mid interaction is not estimated, typically because the event lacks sufficient treated-bound variation in the medium-run window.

Appendix Table C summarizes event support for the event-specific medium-run coefficients used in Figure 3. Of the 156 minimum-wage policy events in the stacked sample, 145 contribute an estimated post-mid event-specific treatment effect. Eleven events do not contribute an estimate because the relevant post-mid interaction is not identified, typically due to insufficient treated-bound variation within the medium-run window or collinearity with the fixed-effects structure. Importantly, these dropped events are few in number and are not disproportionately concentrated in the large-increase tercile: 5 are from the small-increase group, 3 from the medium-increase group, and 3 from the large-increase group.